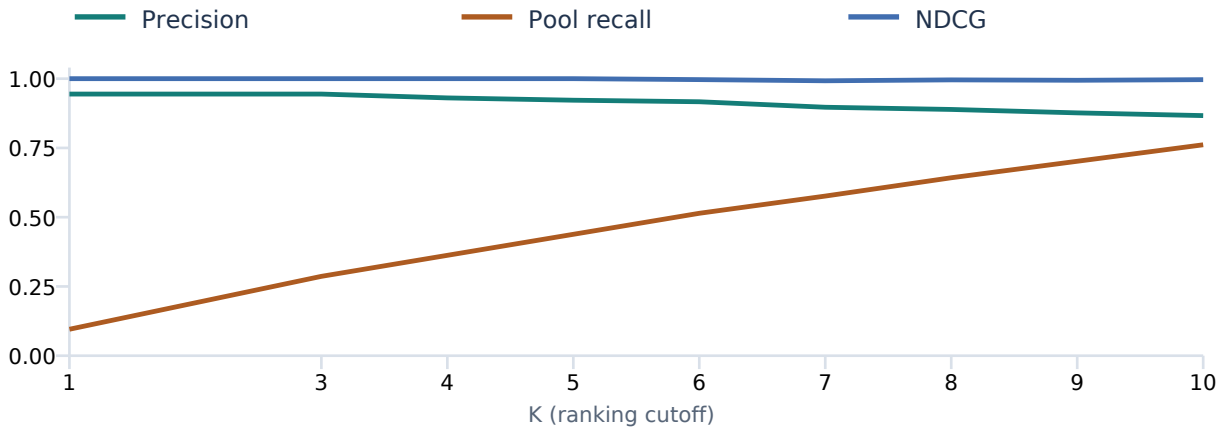
Offline Recommendation Evaluation at K = 1-10

23 October 2026 | Frozen main d5af7e1 | Rental snapshot 9 October 2026

The additional cutoffs show that the sixth position is also useful: **16 of 17** present sixth-ranked results are relevant. Moving from K=5 to K=6 raises judged-pool recall from **43.82% to 51.41%**, while precision changes from **92.22% to 91.67%**. Compare five and six as default display options before validating a production change.

K	Precision@K	Pool Recall@K	NDCG@K	Positions / slots
1	94.44%	9.55%	1.0000	18 / 18
3	94.44%	28.64%	1.0000	54 / 54
4	93.06%	36.23%	1.0000	71 / 72
5	92.22%	43.82%	1.0000	88 / 90
6	91.67%	51.41%	0.9966	105 / 108
7	89.68%	57.62%	0.9924	122 / 126
8	88.89%	64.23%	0.9956	139 / 144
9	87.65%	70.18%	0.9942	156 / 162
10	86.67%	76.14%	0.9964	173 / 180

Macro means: Precision and NDCG use 18 primary scenarios; Recall uses 17. S03 has no relevant pool label. S19-S20 are controls and remain outside all means. All nine listed cutoffs use the same expanded labels; K=2 is outside this revision.



What changed

Revision 3 adds **K=6, 7, 8 and 9**. All 577 judgments, 431 distinct listing IDs, 20 scenarios and saved rankings are unchanged from revision 2. Each scenario still has 25-31 candidates. The earlier five-cutoff results reconcile exactly; no new relevance judgment was needed.

Scope: K=5-10 are offline ranking analyses. The production response limit remains four. No new service execution, model-provider request or production change was made.

Status: preliminary ranking analysis complete; independent human relevance validation and end-to-end acceptance remain pending.

01 / COMPARABILITY AND METHOD

One pool for every cutoff

The original 25 candidates per scenario were retained and every missing native top-ten listing was added. S13/S14 and S15/S16 also share the union of their additions, preserving identical candidate membership within each priority-reversal pair. No candidate was removed and no result was reranked.

K	Original pool recall	Expanded pool recall	Change (percentage points)
1	13.81%	9.55%	-4.27
3	41.44%	28.64%	-12.80
4	53.29%	36.23%	-17.06

Revision 2 expanded the denominator by 74 grade-2/3 judgments. Revision 3 retains that same pool and all labels. The lower restated Recall@1/@3/@4 is not a ranking change; Precision and NDCG still match revision 1. Compare K values on the common expanded pool.

Scoring and provenance

Grades combine mandatory-condition checks with qualitative judgments from **one language model assistant**; **no human rating round has been completed**. The same assistant prepared the benchmark and knows earlier results. Added judgments were assigned from listing facts and the original rubric/anchors, without native scores or ranks in the scoring view, then frozen before metric calculation. Top-ten selection was known, so this is not a blind or independent assessment.

Grade	Meaning	Original	Added	Total
0	Known mandatory-condition failure	135	0	135
1	Hard-eligible, weak preference fit	149	3	152
2	Relevant, with a material preference trade-off	74	34	108
3	Strong preference fit	70	40	110
U	Mandatory evidence unresolved	72	0	72
Total	Scenario-listing judgments	500	77	577

Metric definitions

Precision@K = grade-2/3 hits / K. Empty slots get zero credit.
Pool Recall@K = those hits / all grade-2/3 labels in that scenario's expanded pool. A zero denominator is N/A.
NDCG@K = discounted gain / ideal discounted gain from numeric pool labels, using gain 2^grade - 1 and discount log2(rank + 1). Saved native order, including listing-ID tie breaks, is retained.

U labels remain unknown rather than negatives. They are excluded from Recall and ideal-DCG denominators; any U/unjudged top-K item would make that case/K unavailable. None occurs in the 173 top-ten positions. Results remain conditional on this partially judged, output-enriched pool, not the 21,431-row database.

Precision by scenario

Percentages. All 20 scenarios and nine listed cutoffs use the same frozen ranking and expanded judgments. CSVs and the workbook retain unrounded values, denominators and reasons for N/A.

Case	P@1	P@3	P@4	P@5	P@6	P@7	P@8	P@9	P@10
S01	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S02	100.00	100.00	100.00	100.00	100.00	85.71	75.00	66.67	60.00
S03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
S04	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S05	100.00	100.00	100.00	100.00	100.00	100.00	100.00	88.89	80.00
S06	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S07	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S08	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S09	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S10	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S11	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S12	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S13	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S14	100.00	100.00	100.00	100.00	100.00	85.71	87.50	88.89	90.00
S15	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S16	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S17	100.00	100.00	75.00	60.00	50.00	42.86	37.50	33.33	30.00
S18	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
S20	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Relevant choices and unfilled positions

Grades 2 and 3 count as hits. S17 has only three verified matches: fixed-K Precision is 60% at K=5, 50% at K=6 and 30% at K=10. All other primary cases keep the same Precision between K=5 and K=6. The small macro decline over this step is entirely due to the short list, not a new loss of fit.

At K=10, S02 has six hits, S03 none, S05 eight and S14 nine. Their non-hit positions have grade 1. Controls S19-S20 are not counted in the means.

Judged-pool recall by scenario

Percentages. All 20 scenarios and nine listed cutoffs use the same frozen ranking and expanded judgments. CSVs and the workbook retain unrounded values, denominators and reasons for N/A.

Case	R@1	R@3	R@4	R@5	R@6	R@7	R@8	R@9	R@10
S01	5.88	17.65	23.53	29.41	35.29	41.18	47.06	52.94	58.82
S02	16.67	50.00	66.67	83.33	100.00	100.00	100.00	100.00	100.00
S03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
S04	5.00	15.00	20.00	25.00	30.00	35.00	40.00	45.00	50.00
S05	11.11	33.33	44.44	55.56	66.67	77.78	88.89	88.89	88.89
S06	6.67	20.00	26.67	33.33	40.00	46.67	53.33	60.00	66.67
S07	6.67	20.00	26.67	33.33	40.00	46.67	53.33	60.00	66.67
S08	9.09	27.27	36.36	45.45	54.55	63.64	72.73	81.82	90.91
S09	7.14	21.43	28.57	35.71	42.86	50.00	57.14	64.29	71.43
S10	5.88	17.65	23.53	29.41	35.29	41.18	47.06	52.94	58.82
S11	9.09	27.27	36.36	45.45	54.55	63.64	72.73	81.82	90.91
S12	7.14	21.43	28.57	35.71	42.86	50.00	57.14	64.29	71.43
S13	7.14	21.43	28.57	35.71	42.86	50.00	57.14	64.29	71.43
S14	6.67	20.00	26.67	33.33	40.00	40.00	46.67	53.33	60.00
S15	7.69	23.08	30.77	38.46	46.15	53.85	61.54	69.23	76.92
S16	7.14	21.43	28.57	35.71	42.86	50.00	57.14	64.29	71.43
S17	33.33	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
S18	10.00	30.00	40.00	50.00	60.00	70.00	80.00	90.00	100.00
S19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
S20	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Recall uses a fixed judged-pool denominator

For each scenario, the denominator is the number of grade-2/3 labels in its expanded pool and is identical at every K. Recall is nondecreasing. S03 has zero relevant labels, so its Recall is N/A. S17 reaches 100% at K=3; its remaining empty positions do not increase recall.

Recall excludes U labels and does not cover the entire 21,431-record database. S19 and S20 retain their clarification/no-match control outcomes and remain outside the ranking metric means.

NDCG by scenario

NDCG measures graded ordering against the best sequence available in the judged pool. A value of 1 does not establish user satisfaction. Pool size and relevant-label counts stay fixed across all K values.

Case	N@1	N@3	N@4	N@5	N@6	N@7	N@8	N@9	N@10
S01	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S02	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S03	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S04	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S05	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9688	0.9692
S06	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S07	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S08	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S09	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S13	1.0000	1.0000	1.0000	1.0000	0.9384	0.9410	0.9935	0.9937	0.9939
S14	1.0000	1.0000	1.0000	1.0000	1.0000	0.9215	0.9277	0.9328	0.9725
S15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
S19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
S20	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Where the intermediate cutoffs expose a gap

Losses first appear at K=6 in S13, K=7 in S14 and K=9 in S05. NDCG can recover at a later cutoff because both observed gain and ideal gain change; it need not be monotonic. At K=10 these cases reach 0.9939, 0.9725 and 0.9692 respectively. S03 remains a ceiling case despite weak relevance throughout.

All 173 analyzed positions have numeric grades and pass the declared hard requirements in a static recheck. Overall pool coverage is 505 / 577 numeric judgments; the remaining 72 U judgments are preserved. High NDCG may be optimistic because the pools contain native outputs and the ratings are not independently human-validated.

05 / DEFAULT DISPLAY SIZE

Compare five and six first

Extension	Extra positions	Relevant / weak	Marginal precision
4 to 5	17	16 / 1	94.12%
5 to 6	17	16 / 1	94.12%
6 to 7	17	14 / 3	82.35%
7 to 8	17	15 / 2	88.24%
8 to 9	17	14 / 3	82.35%
9 to 10	17	14 / 3	82.35%

The sixth result has a strong marginal contribution

The fifth and sixth positions each add 16 relevant results out of 17 present positions. Moving from K=5 to K=6 raises macro Recall by 7.59 percentage points while Precision falls by 0.56 points under the fixed-K convention. The NDCG change from 1.0000 to 0.9966 reflects the first graded-ordering gap in S13.

More weak fits appear from the seventh position

The seventh position adds 14 relevant choices and three weak fits; the eighth adds 15 and two; the ninth and tenth each add 14 and three. These positions still contribute coverage, but their marginal relevance is lower than that of positions five and six. Across K=4 to K=10, the additional 102 positions contain 89 relevant and 13 weak-fit results.

Provisional product recommendation

Treat **five and six** as candidate default display sizes, prioritizing validation of six. Keep an optional expansion to ten for users who want more alternatives. Do not fill empty slots by relaxing mandatory criteria. State any unmet soft preference and let users choose whether to explore alternatives.

What remains to establish

This benchmark does not measure user reading effort, click or viewing decisions, satisfaction, latency or live-model cost. It cannot establish a universally optimal K. Independent human relevance labels and a live comparison of the display options are needed. The current interface limit of four must be changed and its full flow revalidated before deploying a larger result count.

Case-specific limits and source checks

Weak results concentrate in specific scenarios

At K=10, 17 of the 173 present positions have grade 1: S03 contributes ten, S02 four, S05 two and S14 one. S17 accounts for all seven empty slots. This separates weak matches from a justified lack of additional verified candidates.

S02 and S03: increasing quantity cannot create fit

S02 has only six relevant listings under its larger-area preference; positions 7-10 are smaller, weak-fit listings. Its Recall reaches 100% while Precision falls to 60%. S03 misses the primary D05 preference at every analyzed position; a longer list leaves Precision at zero.

S05 and S14: inspect preference trade-offs

In S05, positions 9-10 are grade 1 while a grade-2 candidate remains elsewhere in the pool; the binary size preference in the native policy does not mirror the rubric's graded tolerance for a modest size shortfall. In S14, position 7 is the SGD 3,000 near-campus listing 500262756, graded 1 under the price-first scenario, ahead of cheaper grade-3 alternatives. These are policy-versus-rubric discrepancies to review, not confirmed model failures.

Priority reversals and data quality

S13 puts two grade-2 listings before a grade-3 listing at position 8, producing a small NDCG loss. S15 and S16 still have identical top-ten IDs and order: their leading candidates meet both district and size preferences, so this pair still provides little discrimination. Source-review flags now affect 13 analyzed positions and seven distinct IDs, including the HDB record reporting 13,896 sqft and several multi-bedroom records reporting under 200 sqft.

Controls remain separate

S19 retains needs_clarification because mandatory furnishing and six-month lease evidence is unresolved. S20 retains no_match under its tight mandatory limits. Increasing K supplies no new verified result for either control. Their saved outcomes are preserved; no new service execution was performed.

Preserved evidence, explicit limits

Check	Verified result
Source and snapshot	Commit d5af7e1b58b2e7064bf781f2a7a2eb19afc6de57; 53 source files and the original SQLite SHA256 reverified.
Original evidence	All 193 revision-2 files remain recoverable: 24 publication/calculation files archived under history/v2 and 169 shared evidence/input files preserved. Revision 1 still retains 154 files.
Order and judgments	All 20 top-ten extracts match the saved full rankings; their first four match the original service responses. All 500 original rating records are unchanged.
Independent arithmetic	540 case/metric cells, including controls and N/A, and 27 macro results reconciled with a separate NumPy implementation. All 100 prior case/K rows are unchanged.
Coverage and hard criteria	173 / 173 analyzed positions have numeric grades and pass static hard-condition checks. No final gate for K>4 was executed.
New calls and changes	0 service runs, 0 external model-provider calls, no production code changes.

Package map

Group42_Offline_Recommendation_Evaluation_EN.pdf: this updated report.
Group42_K_Comparison_EN.xlsx: nine-cutoff comparison, 180 case/K rows and all 577 ratings.
ratings/ and **inputs/**: unchanged labels, reasons, source facts and human review template.
metrics/: all metrics, top-ten IDs/grades, coverage, reconciliation and marginal results.
raw/ and **history/v1/**: unchanged native evidence and complete original traces.
history/v2/: previous publication and calculation files. Revision_Manifest.json maps all original archived/shared paths and hashes.
reproduction/: calculator, explicit judgments and independent verifier.

Reproduction and boundaries

START_HERE_EN.txt provides commands. Recalculation uses the saved rankings and frozen labels without a model or database. The exact source and database hashes remain in the manifest. The 1 GB database is not duplicated. The native service accepts at most four results; K=5-10 are cutoffs in saved internal rankings, not tested larger service responses.

Acceptance remains pending

Ratings still require independent human validation. Output-driven pool additions limit generalization; this is not full-database Recall or a held-out study. Source unit/scope checks, natural-language intake, live-model orchestration, browser behavior, live database integration and final project acceptance remain outside this result. No new GitHub publication is included.

References: pinned project source; NDCG definition reference (implementation here uses explicit exponential gains); frozen NUS reference, OpenStreetMap contributors, ODbL 1.0.